RE07 Global Infrastructure Investment and Management

7. Technology and Asset Life

Helen Bao
Department of Land Economy, University of Cambridge

Lecture Outline

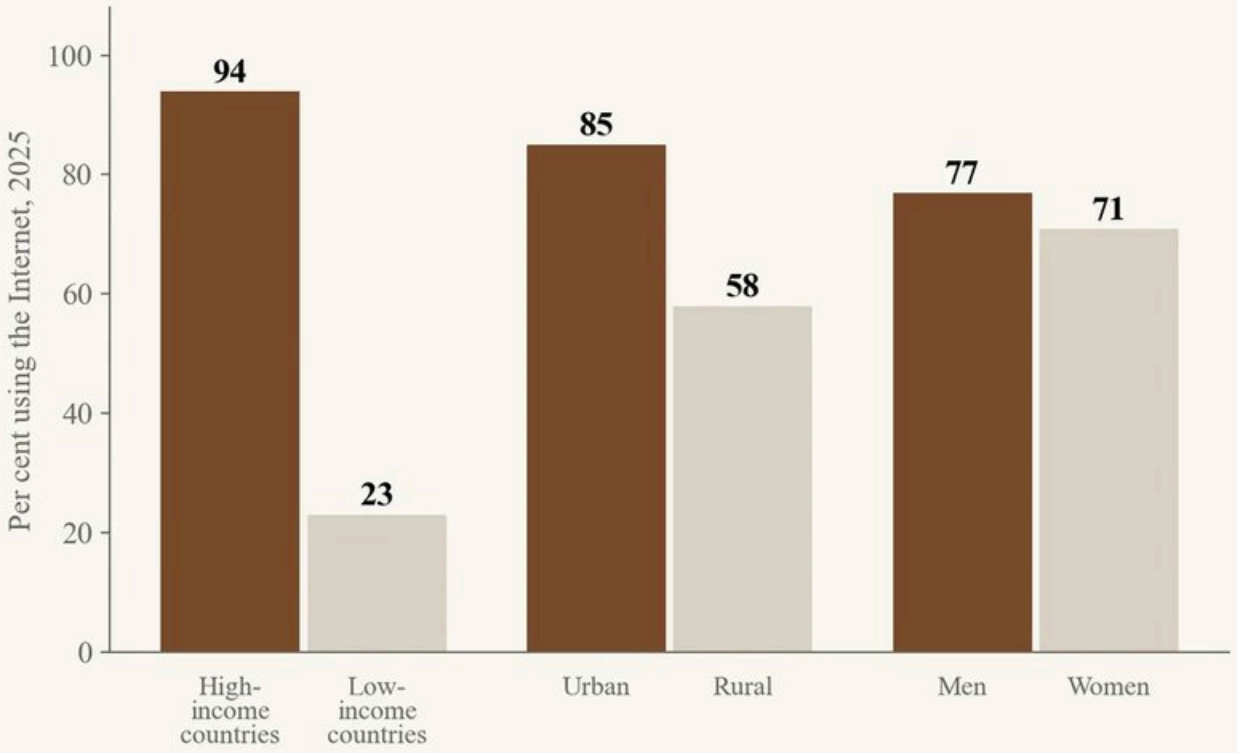
Digital networks as the running case: six layers
Why the demand is real: connectivity and growth
Fibre, towers, cables and satellites: how each earns and how long it lasts
Data centres in the stack
Security, sovereignty and resilience
How investors hold digital assets
Three models of demand, technology and asset life
Case Study: A Digital Infrastructure Platform (Guest to Be Confirmed)

The Running Case: Six Layers of Digital Infrastructure

Layer	What is built	How it earns	A figure from the evidence
Subsea cables	Fibre pairs between landing stations on different coasts	Sale or exchange of capacity; the main buyers are now content and cloud firms	Cables worth over US$14 billion may enter service in 2025 to 2027
Terrestrial fibre	Long-haul and city routes, and the last mile to homes and offices	Wholesale access for retail providers, under price regulation in the UK	Full fibre passes 78% of UK homes; 42% of those have taken it
Towers and masts	Sites with a structure, land and power for operators' antennas	Rent per tenant under long contracts with mobile operators	Tenants per tower range from 1.47 to 2.26 across five listed companies
Satellites	Constellations in low orbit, with ground stations and user terminals	Subscriptions to users and capacity to operators	Over 110,000 Starlink connections in the UK; design life five years or more
Data centres	Buildings with power, cooling and connections, let to cloud and other tenants	Rent per kilowatt of capacity, on leases of several years	Two operators each run over 30 million square feet
Interconnection	Exchanges and carrier-dense sites where networks meet	Port and cross-connection fees	The largest exchanges each exceed 14 Tbps at peak

TeleGeography (2026), State of the Network 2026; Ofcom (2025), Connected Nations 2025; ITU (2025); company results releases of 2025 and 2026; SpaceX. Page references are in the notes.

Connectivity and Growth: The Evidence on Demand



The base: about 6 billion people, three quarters of the world, were online in 2025; 2.2 billion were not, 96 per cent of them in low- and middle-income countries

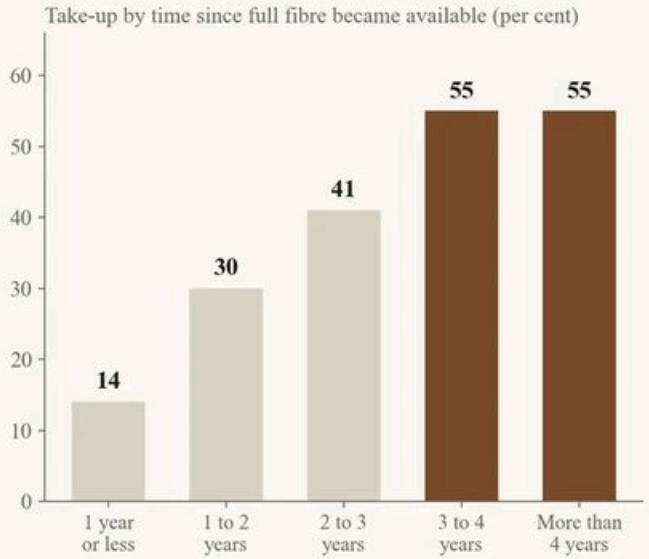
Growth: a ten point rise in broadband penetration raised annual growth in income per head by 0.9 to 1.5 points in OECD countries from 1996 to 2007

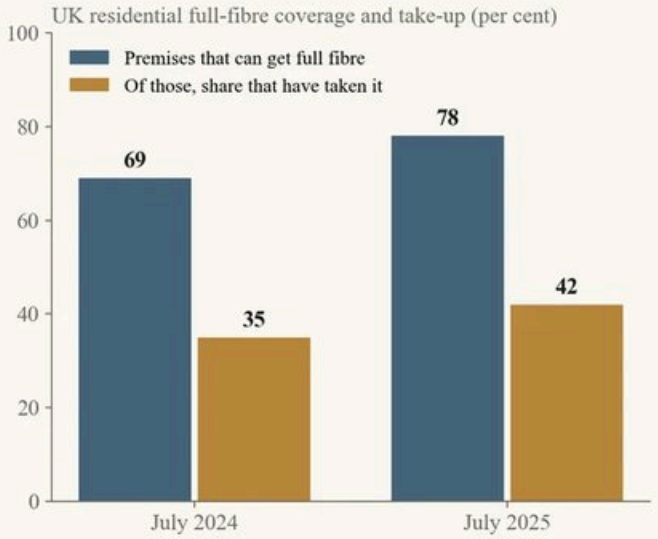
Jobs: the arrival of submarine cables raised the probability of employment in Africa by 3 to 13 per cent in three datasets, a result a 2024 comment says is not robust

The cost: connecting the remaining three billion by 2030 was put at US$428 billion in 2020, most of it for infrastructure

ITU (2025), Facts and Figures 2025; Czernich et al. (2011), Economic Journal 121(552); Hjort and Poulsen (2018), NBER Working Paper 23582; Roodman (2024), arXiv 2401.13694; ITU (2020), Connecting Humanity, page 12.

Fibre to the Home: Coverage Runs Ahead of Take-Up





Coverage: full fibre reached 78 per cent of UK homes in July 2025, up from 69 per cent a year earlier, and gigabit-capable networks 87 per cent

Take-up: 42 per cent of premises that can get full fibre have taken it, 33 per cent of all premises, with rural take-up of 56 per cent above urban at 40

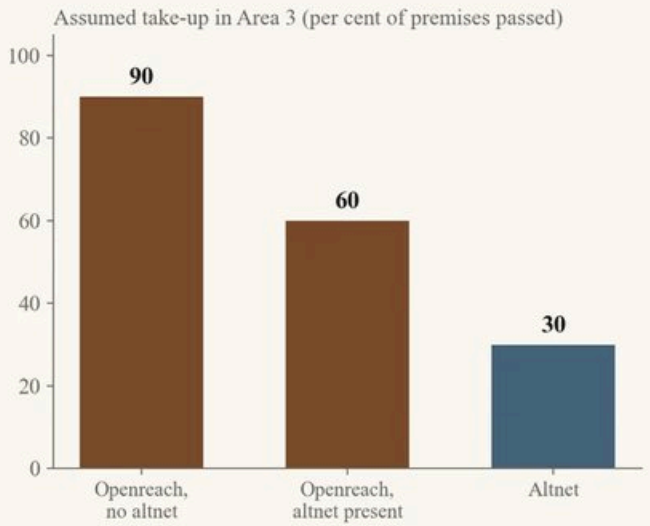
Time: take-up rises with the time since build, from 14 per cent in the first year to 55 per cent after three

The builders: Openreach had passed 20 million premises when Ofcom decided in March 2026 and plans 30 million by 2030, while rival networks build alongside it

Ofcom (2025), Connected Nations 2025, UK report, Tables 2.5 and 2.6 and pages 4, 6, 11 and 12; Ofcom (2026), Telecoms Access Review 2026-31, Volume 4, page 21.

Wholesale Access and the Ladder of Investment





The idea: regulators let entrants climb from renting the incumbent's service to renting its ducts to building their own network (Cave, 2006)

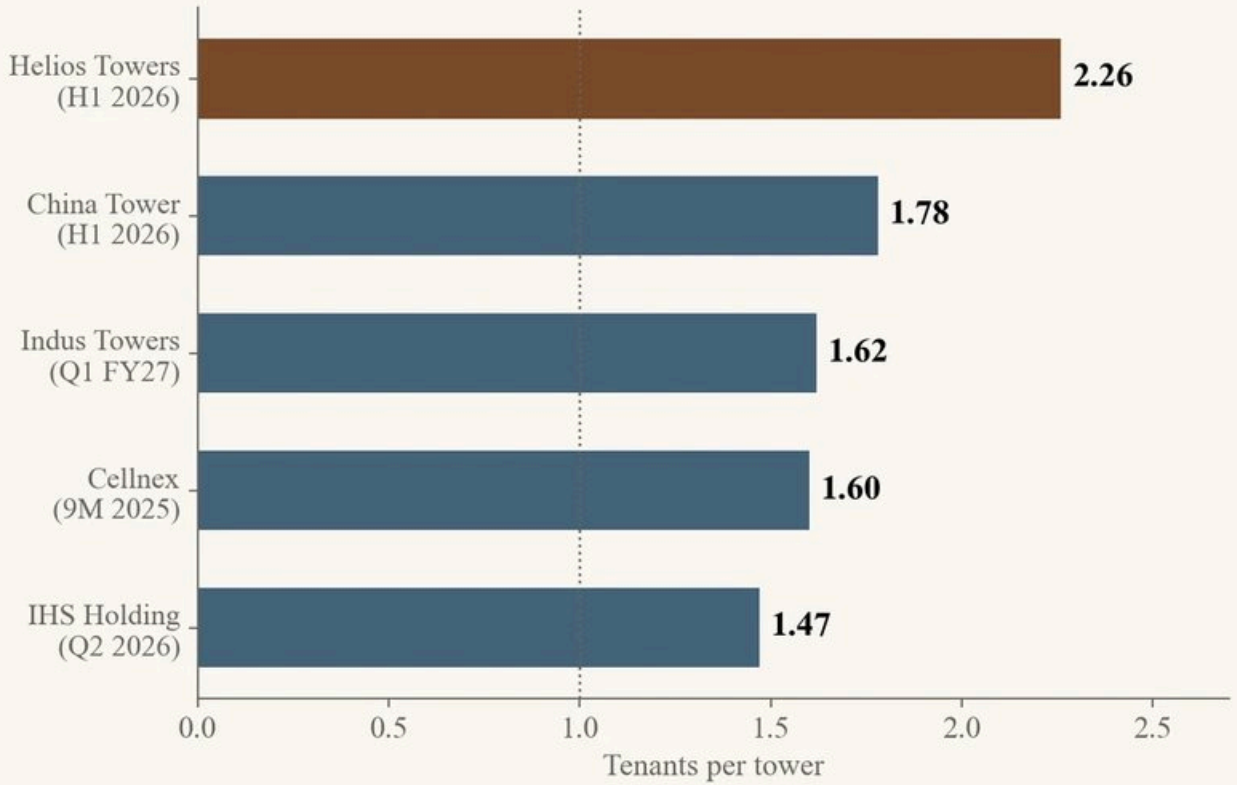
The rule: Openreach must give rivals access to its ducts, and its FTTP 80/20 rental is capped at 18.18 pounds a month for 2026 to 2031

The test: an efficient entrant needs 13.59 to 21.21 pounds a month; the cap sits within that range, towards its upper end

The consequence: a second network divides customers: Ofcom assumes 90 per cent take-up for Openreach alone, 60 per cent with an altnet and 30 for the altnet

Ofcom (2026), Telecoms Access Review 2026-31, Volume 4, paragraphs 1.116 and 1.117, and Annex 12, paragraph A12.35 and footnote 599; Cave (2006), Telecommunications Policy 30(3-4), 223 to 237.

Towers and the Neutral Host



The model: a tower company rents space and power to several mobile operators, so each added tenant brings rent at little extra cost

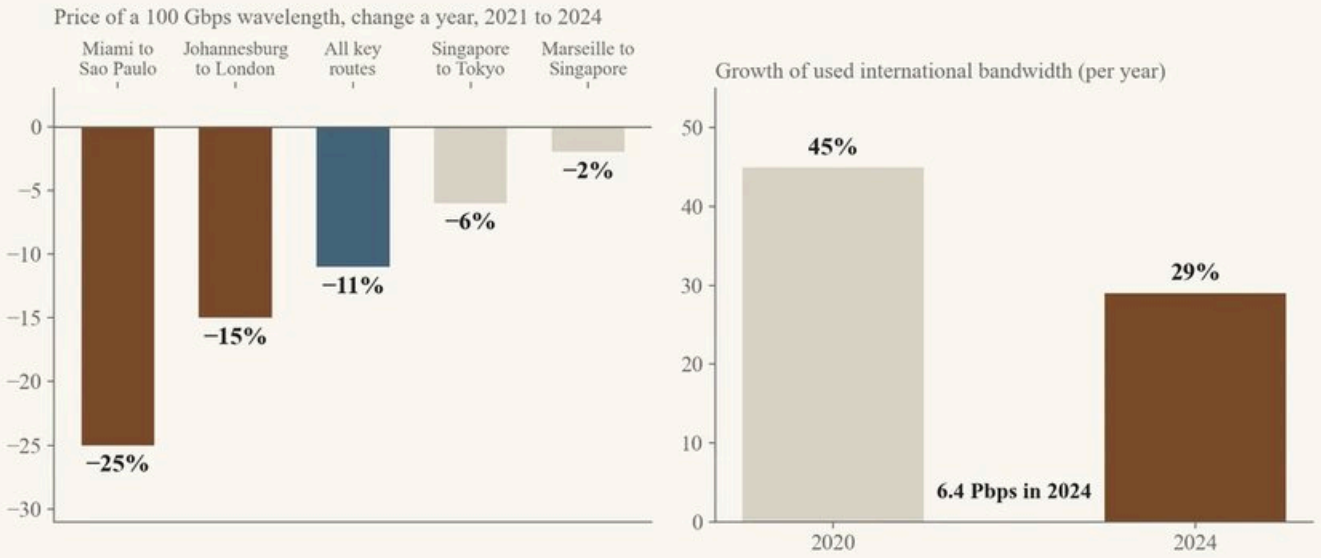
The range: from 2.26 tenants per tower at Helios Towers to 1.47 at IHS Holding; China Tower reports 1.78 on 2.17 million sites

The contracts: Helios has US$5.9 billion of contracted revenue, 98 per cent from large operators, with 6.5 years of initial life remaining on average

The risk: IHS lost 3,704 tenants to churn in a year, 2,576 of them in one quarter when a small Nigerian operator changed its contract

Helios Towers (30 July 2026); China Tower (11 August 2026); Indus Towers (July 2026, as reported); Cellnex (November 2025, February 2026); IHS Holding (Q2 2026). Periods and definitions differ.

Subsea Cables: Content Providers Take the Lead



Scale: international bandwidth demand reached 6.4 petabits a second in 2024, up 29 per cent in the year, slowing from 45 per cent in 2020

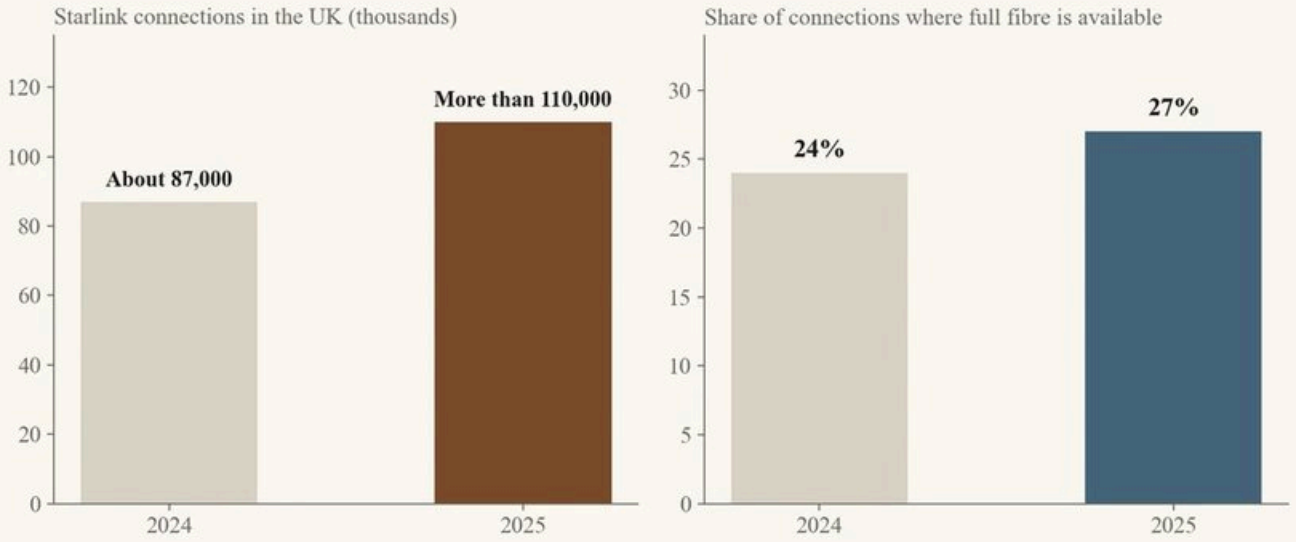
Who uses it: content and cloud networks took almost three quarters of demand, and at least 80 per cent on the three busiest routes

Who pays: cables worth over US$14 billion may enter service in 2025 to 2027; 2Africa was built by a consortium led by Meta

The price: a 100 Gbps wavelength fell 11 per cent a year on average from 2021 to 2024, from 25 per cent on one route to 2 on another

TeleGeography (2026), State of the Network 2026, pages 7 to 15; ITU News (2 February 2026); International Cable Protection Committee (2025); Meta Engineering (17 November 2025).

Satellites: A Complement with a Short Life



The UK: over 110,000 Starlink connections in 2025, from about 87,000; Ofcom has authorised nine non-geostationary licences, seven in low orbit

Where: over 12,600 connections are in places without decent fixed or wireless broadband; 27 per cent are where full fibre is available

The asset: Starlink satellites are designed for five years or more, so the constellation must be replaced continually

The link: a tower company argues that satellites add backhaul demand in rural areas, and cites a forecast that terrestrial networks carry 97 per cent of data in 2040

Ofcom (2025), Connected Nations 2025, UK report, pages 4 and 18; SpaceX, Starlink space safety pages; Helios Towers (2026), H1 results presentation, pages 17 and 28 (company view of an FTI Consulting forecast).

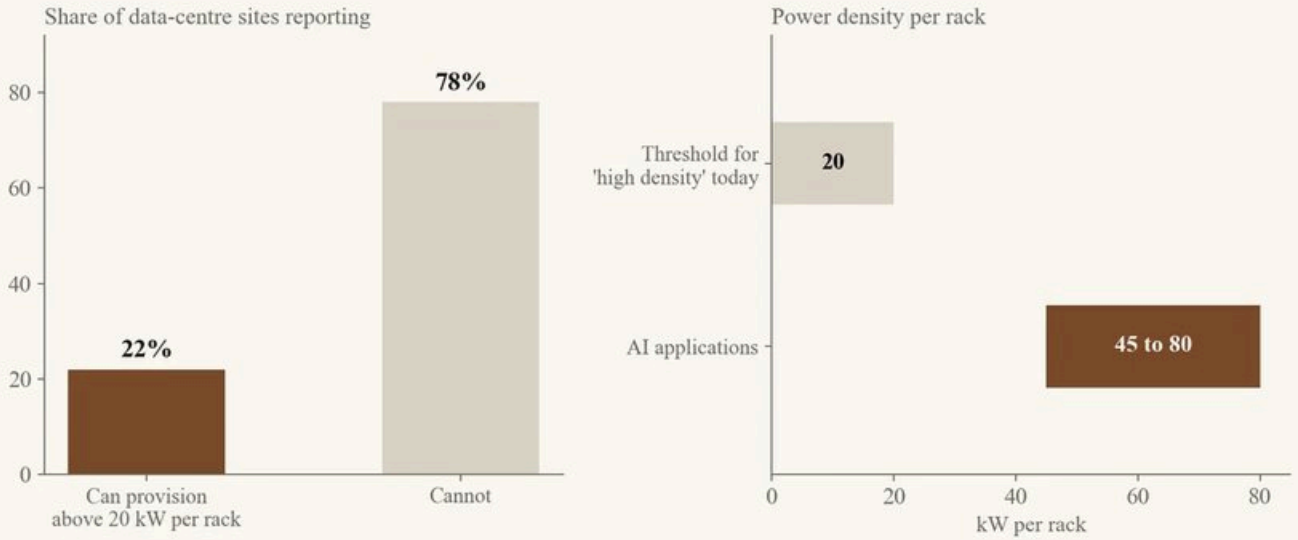
Space Assets and the Infrastructure Test

Applied to orbit, the working definition from Week 1 separates networks that many users rely on from equipment that serves one owner.

Space asset	Network service and asset life	Access and funding
Navigation and timing (GPS, Galileo)	A signal many sectors rely on; the UK benefit was put at £13.6 billion a year	Provided by states, free to users
Geostationary communications	Television, military and weather links; 590 satellites in 2023, designed to last up to 15 years	Protected rights through ITU coordination; no price and no cap on positions
Low-orbit broadband	Over 14,000 operational satellites by end-2025, one operator holding about 60 per cent; Starlink satellites are designed for five years or more	National licences; OneWeb bought out of bankruptcy in 2020 by the UK government and Bharti for $1 billion
Defence communications (Skynet 5)	A government-only network bought as a service	PFI contract signed in 2003, about £2.5 billion over 15 years, extended to 2022
Ground stations and spaceports	Sites, antennas and launch pads that serve the network	The closest fit to conventional investment: EQT's 2024 agreement to buy 80 per cent of Eutelsat's ground stations lapsed in January 2026
A single-customer satellite or a launch vehicle	Equipment: no shared network and no enabling role for other users	Not infrastructure

Wagner (2026), SSRN 7576078; OECD (2026), The Space Economy at a Glance 2026, pages 14 to 15; HSF Kramer (2026), Space infrastructure; London Economics (2021), GOV.UK; IJGlobal and Flightglobal on Skynet 5; press reports on OneWeb. Classification is mine; see notes.

Data Centres: The Building, the Power and the Tenant



The business: operators let space, power and connections; Equinix and Digital Realty each run over 30 million square feet, and over 60 providers run more than a million

The tenants: Amazon, Google, Meta and Microsoft deployed 16 new data centres in a year and have at least 55 more locations in development

The constraint: commercial operators in the top ten markets use about 16 gigawatts; only 22 per cent of sites can supply more than 20 kilowatts a rack, and AI needs 45 to 80

The price: colocation rates of US$400 to US$460 per kilowatt in several hubs, with operators expecting rises of 3 to 5 per cent and power and space the main concern

TeleGeography (2026), State of the Network 2026, Data Centers findings, pages 28 to 30. Commercial operators only; the rates are as printed, without a stated period.

Security, Sovereignty and Resilience

Exposure	Evidence	Response
Physical damage	About 150 to 200 cable faults a year, 70 to 80 per cent from fishing and anchors; cable length up over 70 per cent in a decade, faults flat	Repair vessels and permits; the European Commission set aside EUR 347 million in February 2026 for repair equipment, monitoring and priority projects
Geopolitics	In the Red Sea, disputed permits and attacks on shipping hamper cable laying and repair	Routes chosen around the risk; 13 Cable Projects of European Interest named in 2026
Control	Foreign-adversary ownership of cables and equipment	FCC 25-49: presumption against licences for applicants controlled by a foreign adversary, certification of no Covered List equipment, annual ownership reports
Concentration	Content and cloud firms take almost three quarters of international bandwidth	Buyers build their own routes; regulators watch ownership
Space dependence	Space systems support over half of critical infrastructures and services in OECD countries; reports of navigation signal interference in European air transport rose from 249 in 2021 to 5,624 in 2025	Competitive and resilient markets, traffic co-ordination and debris mitigation, supply-chain mapping (OECD)
Access	Outside China, the Global South is virtually excluded from satellite manufacturing and has marginal operational control	Technology access and transfer, and satellite data-sharing (Rabitz and others)

ICPC media factsheet (2025); ITU News (2 February 2026); European Commission, Blue Economy Observatory (11 February 2026); TeleGeography (2026), pages 10 and 15; FCC 25-49, Federal Register (27 October 2025); OECD (2026), The Space Economy at a Glance 2026, pages 14 to 15 and 85, citing EUROCONTROL (2026); Rabitz, Popovaite and Vilcinskas (2026), Earth System Governance 28, 100323.

How Investors Hold Digital Assets

Layer	How it is held	A transaction	What it shows
Fixed fibre, incumbent	Regulated wholesale network inside an integrated operator	Openreach: fibre to 20 million premises, 30 million planned by 2030	A price cap set against an efficient entrant's cost
Fixed fibre, carve-out	Network sold by the operator to an infrastructure fund	KKR and Telecom Italia's network (FiberCop): EUR 18.8 billion, up to EUR 22 billion with earn-outs	The seller reports it expected to cut its debt by about EUR 14 billion
Towers	Listed platforms; operators sell towers and rent space back	Cellnex sold Ireland for EUR 971 million (February 2025); MTN agreed in February 2026 to buy IHS (about US$6.2 billion)	Both public and private markets trade tower portfolios
Subsea cables	Consortium of operators and content providers	2Africa: Meta with six other owners	Cost and capacity are shared, and the buyers are also the builders
Data centres	Operating platforms, REITs and joint ventures with a tenant	Equinix, Digital Realty; the Hyperion joint venture (Week 4)	A lease to a few tenants is the contract (Weeks 3 and 4)
Ground stations (space)	Ground stations carved out of a satellite operator	EQT agreed in 2024 to buy 80 per cent of Eutelsat's ground stations; cancelled in January 2026	The tower model applied to space; conditions, including French government approval, were not met

Ofcom (2026), Volume 4, page 21; Cellnex (2026), page 5; IHS Holding (2026); reports of the NetCo transaction (KKR, Telecom Italia; reported, not read in the original); Meta Engineering (2025); TeleGeography (2026), page 12; Herbert Smith Freehills Kramer (2026), Space infrastructure: an emerging asset class (EQT and Eutelsat).

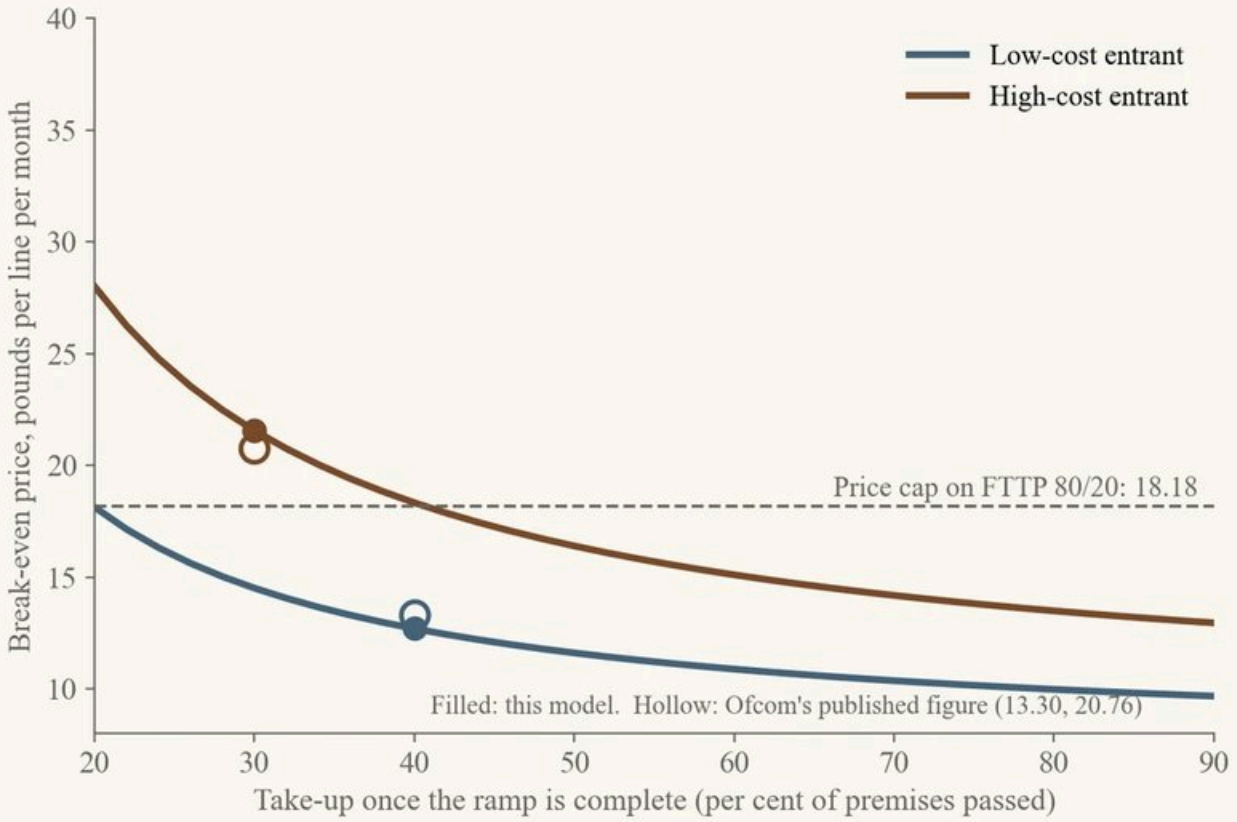
Three Models of Demand, Technology and Asset Life

Model	Layer it describes	Question it answers	Where the numbers come from
1. Roll-out and break-even price	Fixed fibre	How many customers must a network connect before it pays, and what does a second network do to the answer?	Ofcom's Fibre Cost Model, 2026
2. Tenancy and operating leverage	Towers, and any shared passive asset	What does each extra tenant add to the return, and what does a tenant leaving take away?	Helios Towers, first half 2026
3. Falling prices, growing demand and life	Subsea cables, and any capacity asset	What is an asset worth when the price of its output falls every year?	TeleGeography, State of the Network 2026

The three share one structure: a large cost before revenue, revenue that depends on how many use the asset, and a price or a customer base that moves against the owner. Technology enters each differently: through the cost of connecting a customer, through the number of tenants a shared site can hold, and through the price fall that shortens an asset's useful life.

Ofcom (2026), Telecoms Access Review 2026-31; Helios Towers (2026); TeleGeography (2026), State of the Network 2026. Interactive pages in web/.

The Roll-Out Model: Break-Even Price and Take-Up



The structure: a build cost per home passed up front, a connection cost as customers join, a running cost, and a price per connected line per month

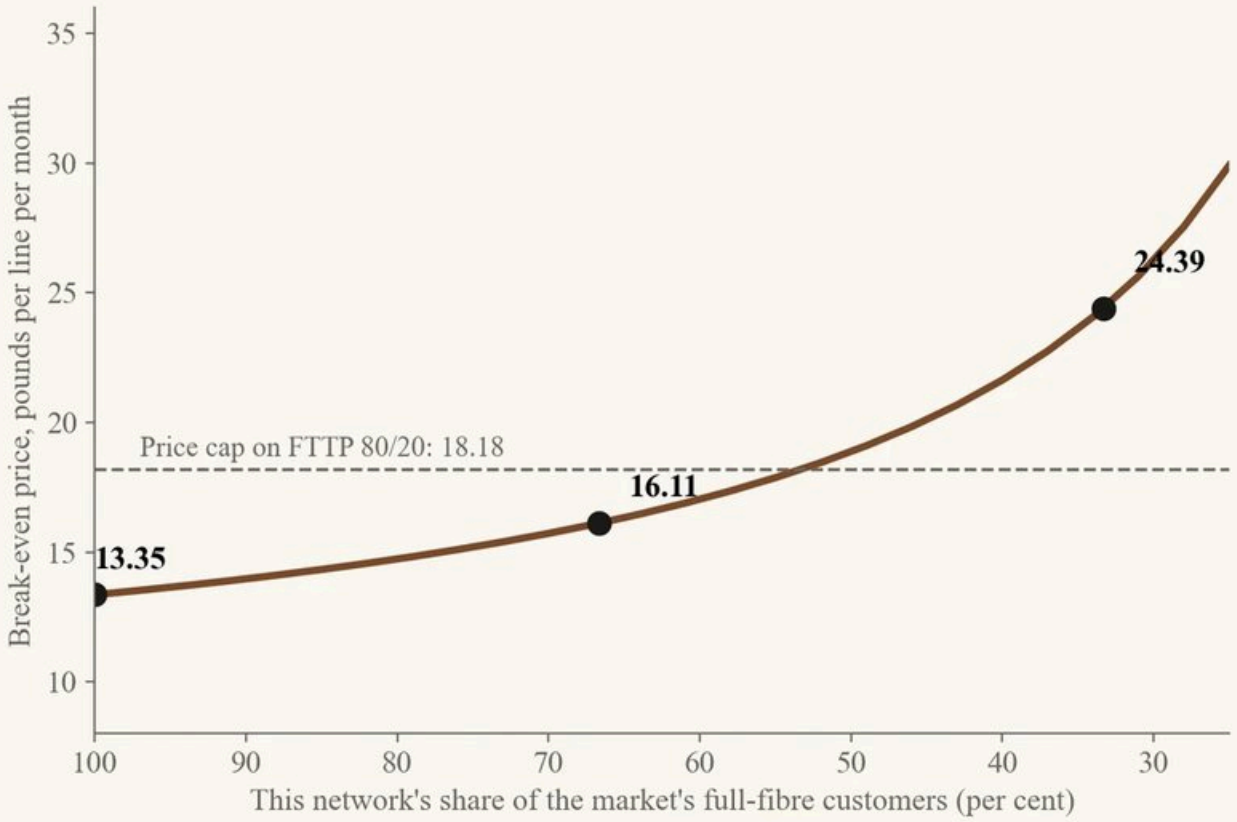
The calibration: on Ofcom's low- and high-cost inputs the break-even price is 12.69 and 21.55 pounds, against Ofcom's published 13.30 and 20.76

The sensitivity: take-up is the swing factor: for the low-cost entrant 11.21 pounds at 55 per cent, 12.69 at 40 and 14.51 at 30

The comparison: the price cap of 18.18 pounds lies inside the range, towards its upper end

Ofcom (2026), Telecoms Access Review 2026-31, Annex 11, Tables A11.5 and A11.6 and paragraph A11.161, page 233; Volume 4, page 20; Ofcom (2025), Connected Nations 2025, page 12. The model is mine.

Break-Even Price, Take-Up and Overbuild



Alone: on Ofcom's central Area 3 costs, a network that has the area to itself needs 13.35 pounds a month, and earns 9.2 per cent a year at the cap price of 18.18

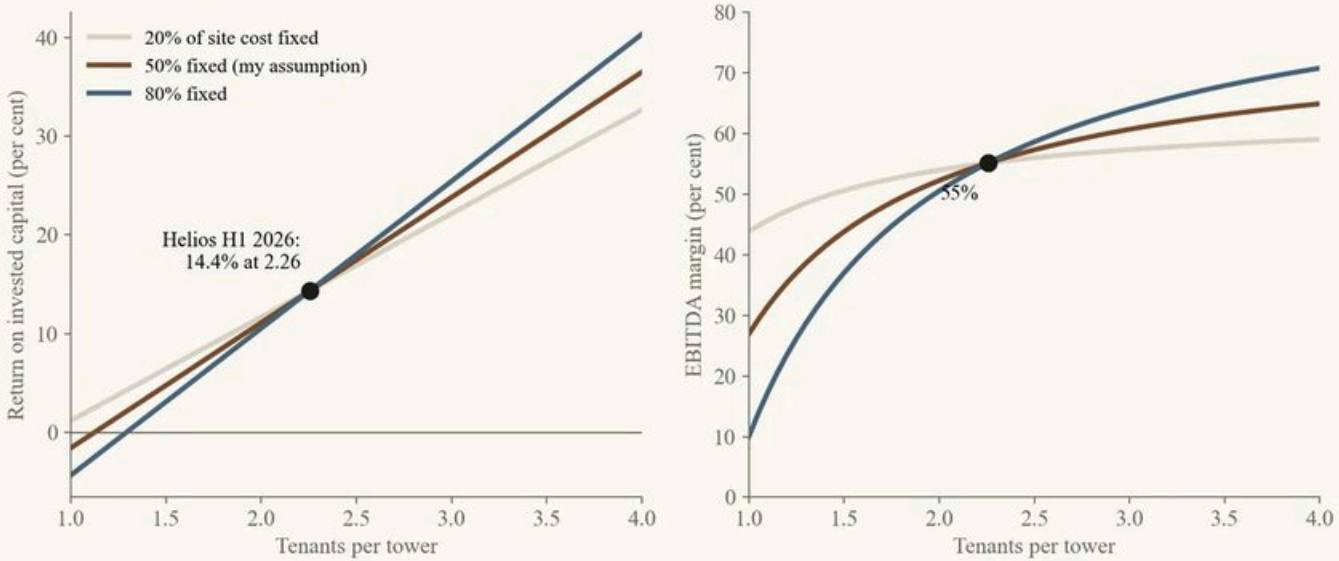
With a second network: keeping two thirds of the customers, it needs 16.11 pounds and earns 6.9 per cent

The rival: a network with a third needs 24.39 pounds and earns 3.3 per cent at the cap price, below its cost of capital

The lesson: a second network divides customers rather than adding them, so the cost of duplication falls on the investors of one or both, or on customers

Ofcom (2026), Telecoms Access Review 2026-31, Annex 12, Tables and paragraphs A12.21, A12.34, A12.35, Table A12.5 and footnote 599 (pages 239 to 243). The model and the real rate are mine. Interactive page in web/.

Tenancy and Operating Leverage



The anchor: 2.26 tenants per tower, an EBITDA margin of 55 per cent and a return on invested capital of 14.4 per cent, from Helios Towers' first half of 2026

The leverage: if half of site cost is fixed, each added tenant brings about US$21 thousand of EBITDA to a site with US$166 thousand of capital, about 13 points of return

The evidence: growth capital of US$316 million in 2023 to 2025 added US$112 million of free cash flow, over 30 per cent, against 14.4 per cent on all capital

The risk: the same arithmetic runs in reverse when a tenant leaves, as IHS found when it lost 3,704 tenants to churn in a year

Helios Towers plc (2026), results for the six months to 30 June 2026, pages 1, 2 and 15; H1 2026 results presentation, pages 7 and 13; IHS Holding (2026). The split of site cost is my assumption.

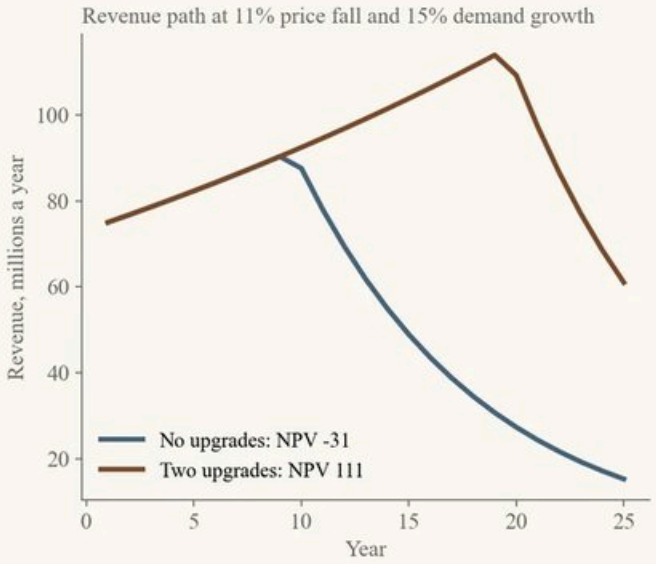
Return on a Tower by Tenancy Ratio

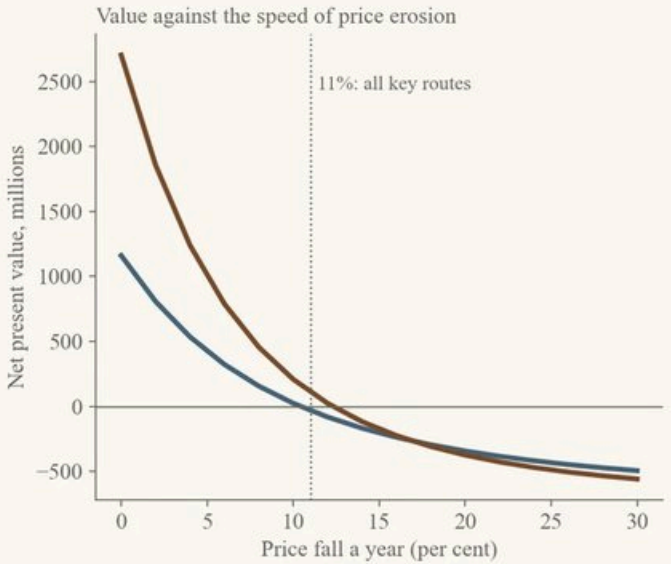
	At 1.5 tenants	At the anchor, 2.26	At 3.0 tenants	Tenants needed for 10%
20% of site cost fixed	6.4%	14.4%	22.2%	1.84
50% of site cost fixed (my assumption)	4.8%	14.4%	23.8%	1.91
80% of site cost fixed	3.1%	14.4%	25.4%	1.96

Return on invested capital of one site, with the anchor's other settings. At the anchor every row gives 14.4 per cent by construction.

Helios Towers plc (2026), pages 1, 2 and 15, annualised; the model and the split of site cost are mine. Interactive page in web/.

Price Decline, Demand Growth and Asset Life





The structure: revenue is price times the smaller of demand and capacity; price falls every year, demand grows, and an upgrade raises capacity at a fraction of the build cost

The data: TeleGeography: route demand up 29 per cent in 2024; the price of a 100 Gbps wavelength down 11 per cent a year, from 2 to 25 by route

The result: with no upgrade revenue peaks in year 9 and falls, and the value is minus 31 million; two upgrades move it to 111 million

The route: at a 2 per cent fall the value with upgrades is 1856 million, at 25 per cent minus 488

TeleGeography (2026), State of the Network 2026, pages 7, 13 and 14. The cost, revenue and upgrade parameters and the model are mine and are not estimates for any cable.

Value of a Capacity Asset by Route and Life

Setting	Value with upgrades (US$ million)	Value with none	Return with upgrades	Revenue peaks in year
Marseille to Singapore: price falls 2% a year	1856	808	21.8%	20
Average of the key routes: 11% a year	111	−31	10.0%	19
Miami to Sao Paulo: 25% a year	−488	−432	none	1
Average route, demand growing at the route rate of 29%	779	123	18.7%	11
Five-year life, 25% fall, no upgrades	−378	−378	−21.8%	1

TeleGeography (2026), State of the Network 2026, pages 13 and 14; the model, the cost and revenue parameters and the upgrade assumptions are mine. Interactive page in web/.

Three Models Compared

	Roll-out (fibre)	Tenancy (towers)	Price decline (cables)
What it asks	What price covers the cost of a network that connects customers over years	What each extra tenant adds to the return of a shared site	What an asset is worth when the price of its output falls and demand grows
What it takes as given	Build and connection cost, take-up, price and the rate	Revenue per tenant, tenants per site, the split of cost fixed and variable	Price fall, demand growth, capacity, upgrade cost and life
On the slides	Break-even 12.69 to 21.55 pounds; 13.35 alone, 24.39 with a third of the customers	14.4 per cent at 2.26 tenants; about 13 points a tenant	Value 111 against minus 31 million with and without upgrades; minus 488 at a 25% fall
What it leaves out	Phasing, churn, competition beyond a share	New-build capital, escalators, tax, the price paid	Contracts, advance sales, faults, rival cables as events
A case could test it by	Take-up by cohort and the price per line	Tenants per site and the cost of adding one	Price per unit by year, capacity sold, upgrade cost

Slides 15 to 20 and the sources named there: Ofcom (2026); Helios Towers (2026); TeleGeography (2026). Interactive pages in web/.

Case Study: A Digital Infrastructure Platform (Guest to Be Confirmed)

The platform and its position: what it owns, by layer and country, and what it depends on

The investments: what was built or bought, for which tenants, and on what contracts

The economics: take-up or letting, tenants per site and how the price per unit has moved

Technology and life: what must be upgraded or replaced, when, and at what cost

Discussion: questions for the guest

Summary

Digital networks as the running case: six layers
Why the demand is real: connectivity and growth
Fibre, towers, cables and satellites: how each earns and how long it lasts
Data centres in the stack
Security, sovereignty and resilience
How investors hold digital assets
Three models of demand, technology and asset life
Case Study: A Digital Infrastructure Platform (Guest to Be Confirmed)